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SVD or PCA for Image Analysis in Medical Diagnostics

Statistical Methods — Project Report

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Contents

1	Problem Description and Motivation	2
1.1	Clinical Motivation	2
1.2	Dataset	2
2	Theoretical Background	3
2.1	Singular Value Decomposition (SVD)	3
2.1.1	Rank- k Approximation	3
2.1.2	Compression Ratio and Quality Metrics	4
2.1.3	Efficient MSE Computation	4
2.2	Principal Component Analysis (PCA)	4
2.2.1	Choosing k : Percentage of Variance Explained	5
2.2.2	Eigenfaces / Eigen-X-rays	5
2.3	Relationship between SVD and PCA	5
3	Implementation	5
3.1	Software Architecture	5
3.2	Key Configurable Parameters	6
3.3	Phase 2 — Pre-processing Pipeline	6
3.4	Phase 3 — SVD Engine	7
3.5	Phase 5 — PCA	7
4	Numerical Experiments and Results	8
4.1	Phase 3 — SVD Decomposition Analysis	8
4.2	Phase 4 — Reconstruction Quality vs. Rank k	8
4.3	Phase 5 — Statistical Analysis and PCA	10
4.3.1	Scree Plot and Singular Value Decay	10
4.3.2	MSE and PSNR Curves vs. k	10
4.3.3	PCA — 2D Projection and Class Separability	11
4.3.4	Eigen-X-rays	11
5	Comments and Analysis of Results	12
5.1	SVD for Compression — Key Findings	12
5.2	PCA for Exploratory Analysis — Key Findings	12
5.3	Limitations	12
5.4	Answers to the Research Questions	12
5.5	Possible Extensions	13

1 Problem Description and Motivation

1.1 Clinical Motivation

Chest X-rays are among the most widely used diagnostic tools in modern medicine. Every hospital produces thousands of radiographic images per day, creating two concrete engineering challenges:

1. **Storage.** A high-resolution radiograph can occupy tens of MB. At hospital scale, efficient compression without diagnostic loss is critical.
2. **Telemedicine transmission.** Remote diagnostics requires sending images over bandwidth-limited networks, demanding lightweight yet faithful representations.

This project addresses two research questions:

1. Can Singular Value Decomposition compress chest X-rays while preserving diagnostically relevant information?
2. Can Principal Component Analysis reveal latent structure in the dataset that distinguishes different pulmonary pathologies?

1.2 Dataset

The dataset consists of grayscale chest X-rays divided into **four diagnostic classes**:

Class	Pathology	Available	Used
Corona Virus Disease	COVID-19 pneumonia	122	≤ 80
Normal	Healthy lungs	540	≤ 80
Pneumonia	Bacterial/viral pneu.	186	≤ 80
Tuberculosis	Pulmonary tuberculosis	333	≤ 80

Table 1: Dataset composition. Images are capped at 80 per class to balance computation time.

Remark 1 (Imbalanced dataset). The classes have very different cardinalities: Normal contains $\approx 4\times$ more images than COVID-19. This imbalance can bias PCA components toward the Normal class patterns. Comparative conclusions should be interpreted with this limitation in mind.

Raw images have heterogeneous resolutions (from 203×230 px for COVID-19 to 1654×1678 px for Normal). Each image is uniformly resized to 256×256 px using Lanczos interpolation before analysis.

Histogram observations (Exploration Phase)

- **COVID-19:** intensity concentrated in mid-high range (100–170), consistent with ground-glass opacities.
- **Normal:** uniform distribution across the full range, good tissue contrast.
- **Pneumonia:** strong peak in dark tones (30–80), consistent with dense pulmonary consolidations.
- **Tuberculosis:** bimodal distribution with peak in bright tones (150–190), indicative of fibrotic/calcified lesions.

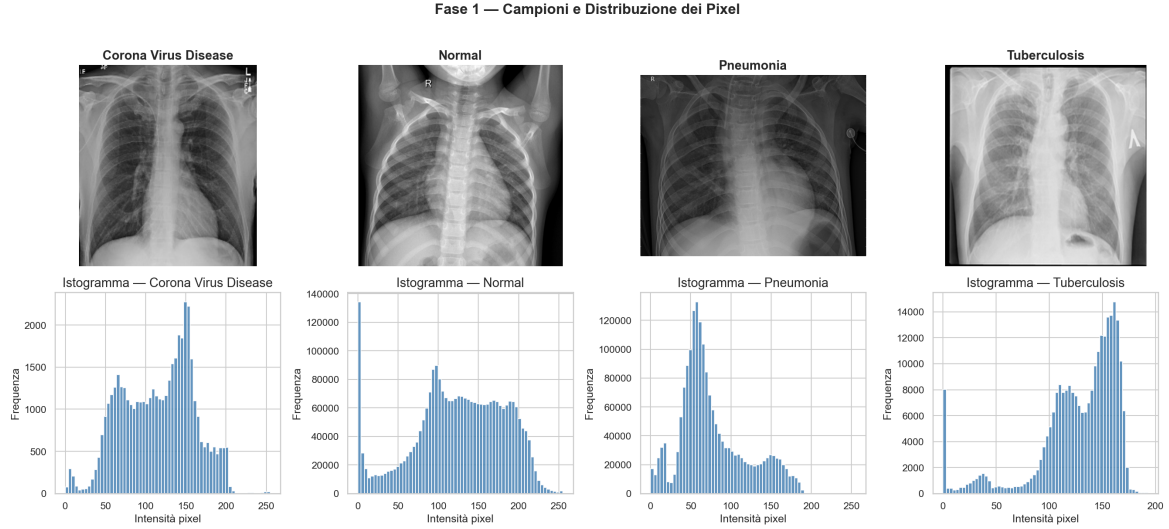


Figure 1: Phase 1 — Sample radiographs and pixel intensity histograms for each class.

2 Theoretical Background

2.1 Singular Value Decomposition (SVD)

Definition 1 (SVD). For any matrix $A \in \mathbb{R}^{m \times n}$, there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$, and a non-negative diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$ such that

$$A = U \Sigma V^T,$$

where the diagonal entries $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$ (with $r = \min(m, n)$) are the *singular values* of A .

The columns of U are called *left singular vectors* (capture row-wise spatial structure), while the rows of V^T are *right singular vectors* (capture column-wise structure). The singular values quantify the “importance” of each component: σ_1 encodes the direction of highest energy variation, and subsequent values capture progressively finer patterns.

connection with eigendecomposition. The singular values and vectors satisfy:

$$A^T A v_i = \sigma_i^2 v_i, \quad A A^T u_i = \sigma_i^2 u_i,$$

so the columns of V are eigenvectors of $A^T A$ and the columns of U are eigenvectors of $A A^T$, with $\sigma_i = \sqrt{\lambda_i}$.

2.1.1 Rank- k Approximation

Retaining only the first k components yields the *truncated SVD* (also called economy SVD when $k = r$):

$$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T = U_k \Sigma_k V_k^T.$$

Theorem 1 (Eckart–Young, 1936). Among all matrices of rank at most k , A_k is the closest to A in both the spectral norm and the Frobenius norm:

$$A_k = \arg \min_{\text{rank}(B) \leq k} \|A - B\|_F,$$

with minimum error

$$\|A - A_k\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2}, \quad \|A - A_k\|_2 = \sigma_{k+1}.$$

This theorem provides the theoretical optimality guarantee for SVD-based image compression: *no other rank- k decomposition can approximate the original image more faithfully.*

2.1.2 Compression Ratio and Quality Metrics

With k components, one stores $k(m + n + 1)$ values instead of mn :

$$\text{compression ratio} = \frac{k(m + n + 1)}{mn}.$$

For $m = n = 256$ and $k = 20$: compression ratio $\approx 15.7\%$.

Two standard metrics evaluate reconstruction quality:

Metric	Formula	Threshold (radiology)
MSE	$\frac{1}{mn} \sum_{i,j} (A_{ij} - \hat{A}_{ij})^2$	lower is better
PSNR	$10 \log_{10} \left(\frac{1}{\text{MSE}} \right)$ [dB]	> 30 dB acceptable

Remark 2. Pixel-wise metrics (MSE, PSNR) do not directly capture clinical saliency. A reconstructed image may score PSNR > 30 dB yet exhibit subtle artefacts in diagnostically critical regions (e.g., lesion boundaries). They are nonetheless the engineering standard for compression fidelity assessment.

2.1.3 Efficient MSE Computation

By the Eckart–Young theorem, the MSE at rank k can be computed directly from the singular values, without reconstructing the image:

$$\text{MSE}(k) = \frac{\|A - A_k\|_F^2}{mn} = \frac{\sum_{i=k+1}^r \sigma_i^2}{mn}.$$

This avoids 200 full matrix multiplications when computing the MSE curve over $k = 1, \dots, 200$.

2.2 Principal Component Analysis (PCA)

Definition 2 (PCA). Given a data matrix $X \in \mathbb{R}^{N \times d}$ (each row is an observation), PCA finds an ordered orthonormal basis $\{v_1, v_2, \dots, v_k\}$ such that the projection $X_{\text{PCA}} = X V_k$ has maximum variance in each successive direction.

The standard algorithm proceeds as follows:

1. **Flatten.** Each image (256×256) is unrolled into a vector $\mathbf{x} \in \mathbb{R}^{65536}$.
2. **Stack.** Form data matrix $X \in \mathbb{R}^{N \times 65536}$.
3. **Center.** Subtract the column mean: $\tilde{X} = X - \bar{X}$.
4. **PCA.** Find the top- k eigenvectors of the covariance matrix $C = \frac{1}{N-1} \tilde{X}^T \tilde{X} \in \mathbb{R}^{d \times d}$.
5. **Project.** $X_{\text{PCA}} = \tilde{X} V_k \in \mathbb{R}^{N \times k}$.

2.2.1 Choosing k : Percentage of Variance Explained

The i -th principal component explains a fraction $\lambda_i / \sum_j \lambda_j$ of the total variance. In practice, k is chosen via a *scree plot* (elbow method) or by requiring a cumulative explained variance threshold (e.g., 90% or 95%).

2.2.2 Eigenfaces / Eigen-X-rays

The eigenvectors v_i of C can be reshaped and visualised as 256×256 images. In the context of chest X-rays they encode the dominant modes of visual variability across the dataset (global contrast, lateral symmetry, mediastinal structure, etc.).

2.3 Relationship between SVD and PCA

A crucial theoretical insight is that **PCA is equivalent to the SVD of the centred data matrix**:

$$\tilde{X} = U \Sigma V^T \implies C = \frac{1}{N-1} \tilde{X}^T \tilde{X} = \frac{1}{N-1} V \Sigma^2 V^T.$$

Hence the columns of V are simultaneously the right singular vectors of $\tilde{X}$ and the eigenvectors of C , with eigenvalues $\lambda_i = \sigma_i^2 / (N - 1)$.

	SVD for compression	PCA for analysis
Input	Single image $A \in \mathbb{R}^{m \times n}$	Dataset $X \in \mathbb{R}^{N \times d}$
Output	Approximation A_k	Projection $X_{\text{PCA}} \in \mathbb{R}^{N \times k}$
Goal	Reduce storage footprint	Find shared latent structure
Theoretical guarantee	Eckart–Young optimality	Maximum variance directions
Use case	Compression, transmission	Visualisation, clustering

SVD and PCA are **complementary**, not alternatives. The SVD is applied *per image* to compress it; the PCA is applied to the *entire dataset* to reveal its collective structure.

3 Implementation

3.1 Software Architecture

The project is written in Python 3.12 and organised into independent modules following the Single Responsibility Principle:

```

1 Medical Image Compression and Analysis/
2 |-- config.py           # global parameters (paths, constants, plot style)
3 |-- data_loader.py      # loading, validation, tilt correction (Phases 1-2)
4 |-- svd_engine.py       # SVD, truncated reconstruction, MSE/PSNR (Phase 3)
5 |-- visualization.py    # plots for phases 1, 2, 4
6 |-- src/analysis.py     # scree plot, MSE/PSNR curves, PCA, eigenfaces (Phase
7                           5)
7 |-- main.py             # orchestration script
8 |-- medical_image_svd_pca.ipynb # interactive notebook
9 '-- output/              # saved figures (.png, 150 dpi)

```

Listing 1: Project structure

Core dependencies: `numpy`, `scipy`, `Pillow`, `scikit-learn`, `matplotlib`, `seaborn`.

3.2 Key Configurable Parameters

Parameter	Default	Description
TARGET_SIZE	(256, 256)	Resize resolution
MAX_PER_CLASS	80	Images per class
CORRECT_TILT	True	Enable tilt correction
MAX_TILT_ANGLE	10	Max allowed tilt (degrees)
K_VALUES_DEMO	[1, 5, 10, 20, 50, 100, 200]	k values for Phase 4
K_RANGE_METRICS	range(1, 201)	k range for MSE/PSNR curves
PCA_N_COMPONENTS	50	Number of PCA components

Table 2: Global parameters defined in `config.py`.

3.3 Phase 2 — Pre-processing Pipeline

```

1 # 1. Load as grayscale
2 img = Image.open(path).convert('L')
3
4 # 2. Resize to 256x256 with Lanczos interpolation
5 img = img.resize((256, 256), Image.LANCZOS)
6 arr = np.array(img, dtype=np.float64) / 255.0 # normalise [0,1]
7
8 # 3. Detect and correct tilt
9 if apply_tilt_correction:
10     arr, angle = correct_tilt(arr, angle=angle)
11     arr = np.clip(arr, 0, 1)

```

Listing 2: Core pre-processing pipeline in `data_loader.py`

Tilt Detection Algorithm

Chest X-rays often exhibit slight rotational misalignment. The detection algorithm maximises the variance of the gradient of the horizontal projection:

$$\theta^* = \arg \max_{\theta} \text{Var}(\nabla_{\text{row}}[\text{rotate}(A, -\theta)]) .$$

A two-pass search is used for efficiency: a coarse scan (step 2° , range $\pm 25^\circ$) on a thumbnail image (128 px), followed by a fine scan (step 0.5° , window $\pm 3^\circ$ around the coarse best).

Images are discarded if $|\theta^*| > 10^\circ$ (excessive tilt), if they contain artificial white-padding corners (median brightness $> 130/255$ in three or more corners), or if they are identified as lateral projections (fewer than 60% of columns carry significant signal).

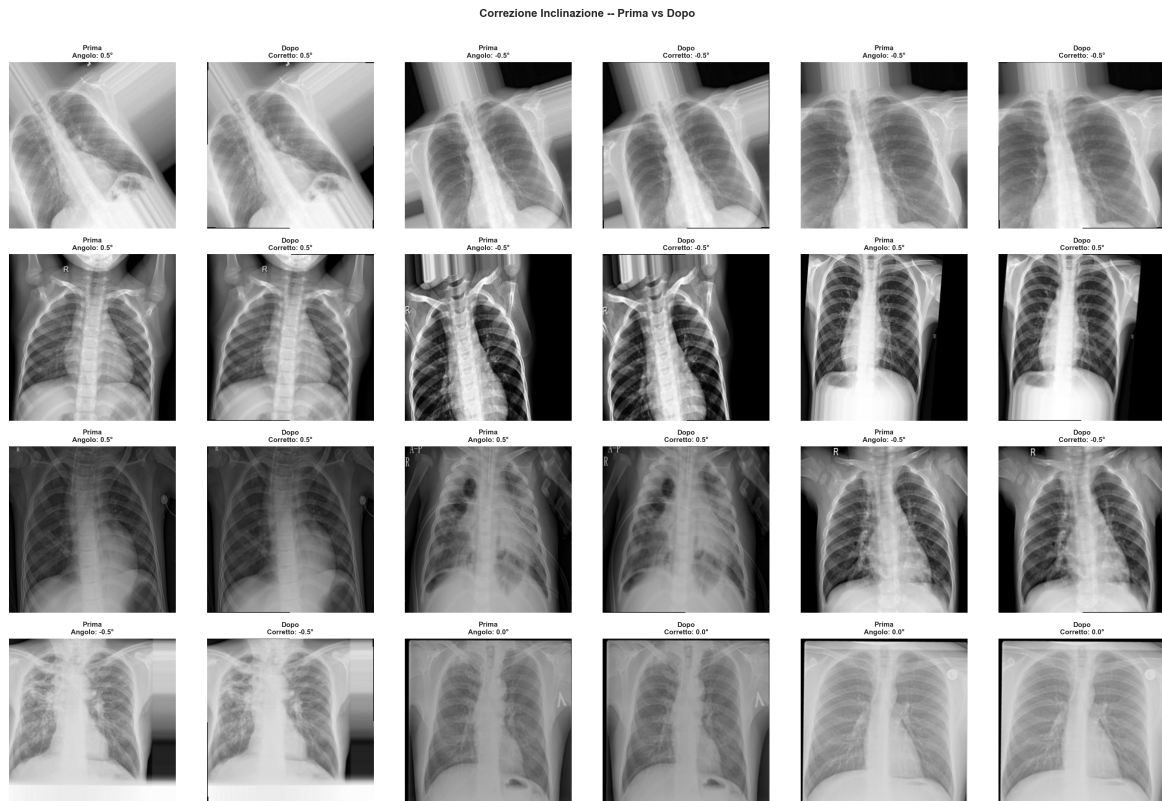


Figure 2: Phase 2 — Examples of tilt detection and correction.

3.4 Phase 3 — SVD Engine

```

1 import numpy as np
2
3 def apply_svd(image_matrix):
4     """Economy SVD of a 2-D image array."""
5     U, S, Vt = np.linalg.svd(image_matrix, full_matrices=False)
6     return U, S, Vt
7
8 def reconstruct_svd(U, S, Vt, k):
9     """Rank-k reconstruction:  $A_k = U_k * \text{diag}(S_k) * Vt_k$ ."""
10    return (U[:, :k] * S[:k]) @ Vt[:k, :]
11
12 def compression_ratio(m, n, k):
13     """Fraction of values stored relative to the full image."""
14     return min(1.0, k * (m + n + 1) / (m * n))

```

Listing 3: SVD core functions in `svd_engine.py`

3.5 Phase 5 — PCA

```

1 from sklearn.decomposition import PCA
2
3 def run_pca(all_images, n_components=50):
4     X = all_images.reshape(all_images.shape[0], -1) # (N, 65536)
5     pca_model = PCA(n_components=n_components)
6     X_pca = pca_model.fit_transform(X) # (N, 50)
7     return pca_model, X_pca

```

Listing 4: PCA execution in `src/analysis.py`

`sklearn`'s PCA uses a randomised SVD internally (`svd_solver='auto'`), making it numerically stable and efficient even for high-dimensional inputs like 65536-feature image vectors.

4 Numerical Experiments and Results

4.1 Phase 3 — SVD Decomposition Analysis

For a 256×256 image (economy SVD), the decomposition produces $r = 256$ singular values. Key observations:

- The *first* singular value $\sigma_1 \approx 130$, capturing global brightness.
- Singular values decay rapidly: after the first 10–15, values are near zero.
- 90% of cumulative variance is captured by approximately $k = 1$ –3 components.

4.2 Phase 4 — Reconstruction Quality vs. Rank k

k	Compr. Ratio	PSNR (dB)	Visual quality
1	0.8%	17.3	Mean brightness only
5	3.9%	23.8	Chest silhouette barely visible
10	7.8%	27.8	Ribs, mediastinum beginning to appear
20	15.7%	32.4	Diagnostic quality (PSNR > 30 dB)
50	39.1%	40.1	Diagnostic detail fully preserved
100	78.3%	50.0	Nearly indistinguishable from original
200	100%	71.5	Virtually perfect reconstruction

Table 3: SVD reconstruction quality as a function of rank k (256×256 Normal class sample, pixel values in $[0,1]$).

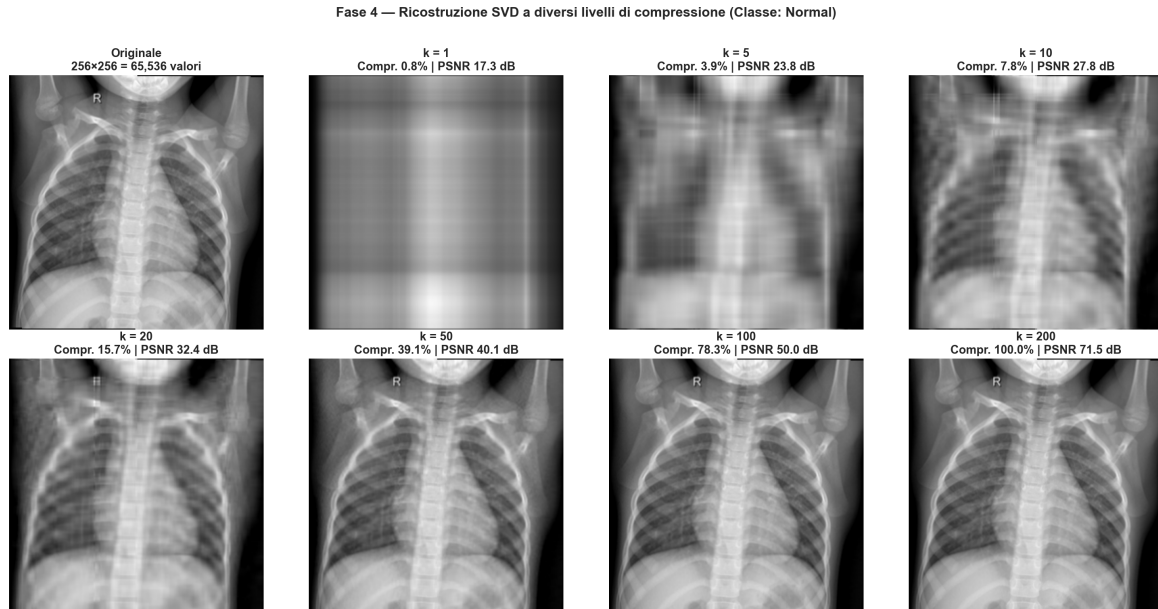


Figure 3: Phase 4 — SVD reconstructions at increasing values of k .

Key result: with only $k = 20$ components (15.7% of the original data), the PSNR exceeds **32 dB** — the conventional threshold for acceptable diagnostic image quality.

Inter-class comparison

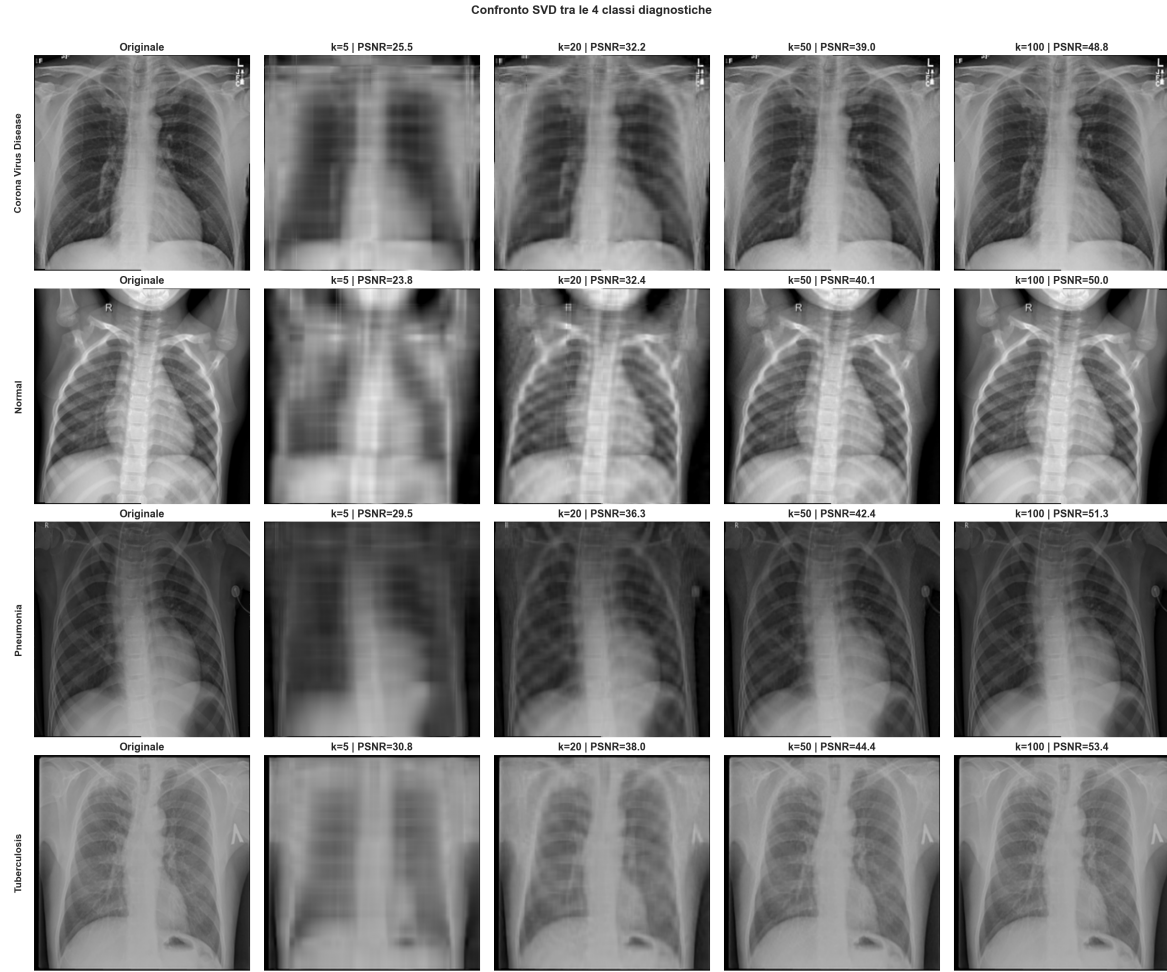


Figure 4: Phase 4 — SVD reconstruction comparison across the four classes at $k \in \{5, 20, 50, 100\}$.

All four classes reach $\text{PSNR} > 30$ dB at $k = 20$ and $\text{PSNR} > 48$ dB at $k = 100$. Pneumonia and Tuberculosis achieve slightly higher PSNR at the same k , indicating lower-rank structure (more compressible). COVID-19 and Normal show marginally higher MSE at low k , consistent with their greater structural complexity.

4.3 Phase 5 — Statistical Analysis and PCA

4.3.1 Scree Plot and Singular Value Decay

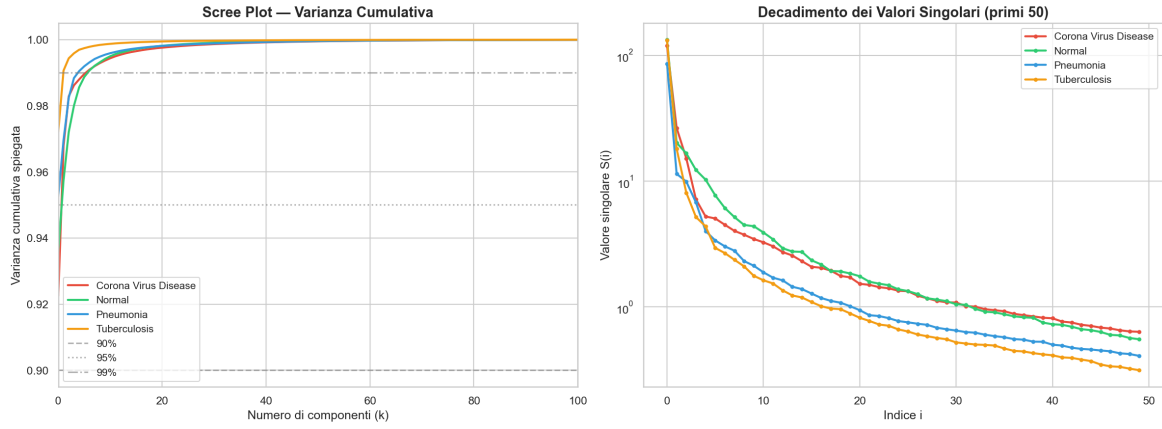


Figure 5: Phase 5 — Cumulative variance explained (left) and singular value decay (right, log scale) for each class.

Class	$k_{90\%}$	$k_{95\%}$	$k_{99\%}$	Total
COVID-19	$\approx 1-3$	$\approx 5-10$	$\approx 20-40$	256
Normal	$\approx 1-3$	$\approx 5-10$	$\approx 20-40$	256
Pneumonia	$\approx 1-3$	$\approx 5-10$	$\approx 20-40$	256
Tuberculosis	$\approx 1-3$	$\approx 5-10$	$\approx 20-40$	256

Table 4: Minimum k required to reach each variance threshold. Results are consistent across all classes.

All classes exhibit a **strong low-rank structure**: 90% of the image energy is concentrated in only 1–3 singular components. This is a direct consequence of the high spatial correlation in chest radiographs (uniform background, repeated curved rib structures).

4.3.2 MSE and PSNR Curves vs. k

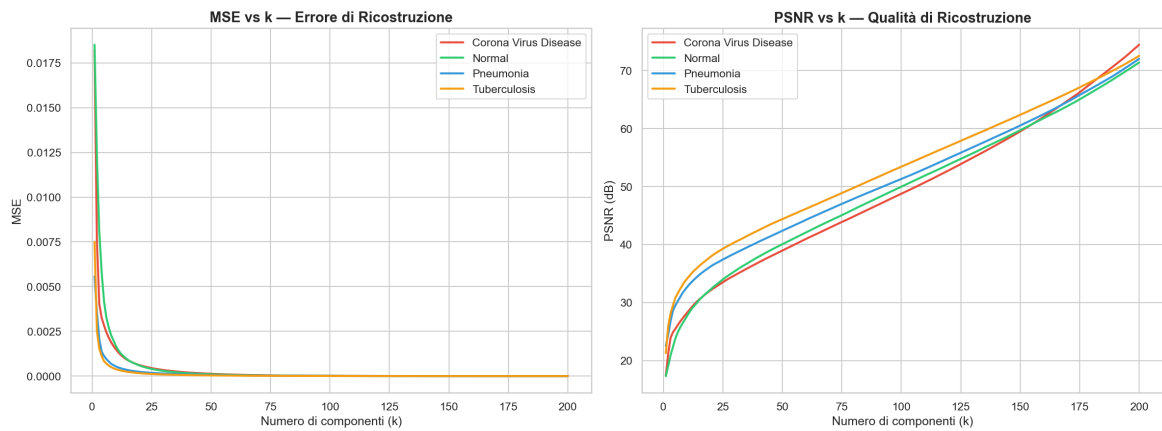


Figure 6: Phase 5 — MSE (left) and PSNR (right) as a function of k for each class.

- **MSE** decays rapidly up to $k \approx 20-30$, then stabilises.

- **PSNR** grows logarithmically; the “elbow” (maximum marginal gain) is at $k \approx 20$ –50.
- COVID-19 and Normal show slightly higher MSE at low k , confirming their greater structural complexity.

4.3.3 PCA — 2D Projection and Class Separability

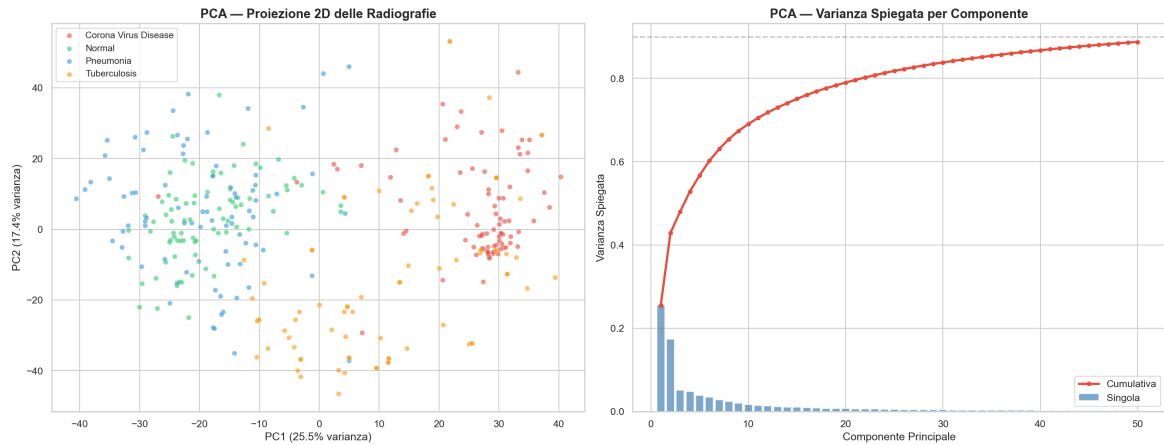


Figure 7: Phase 5 — PCA scatter plot (left) and per-component explained variance (right). The first 50 components explain $\approx 90\%$ of the total variance.

Results:

- PC1 (25.6% variance) and PC2 (17.4%) together explain **43%** of total variance.
- **Normal** and **Pneumonia** cluster on the left ($PC1 < 0$); **COVID-19** and **Tuberculosis** on the right ($PC1 > 0$).
- Considerable class **overlap** remains — partial but not perfect separability.

4.3.4 Eigen-X-rays

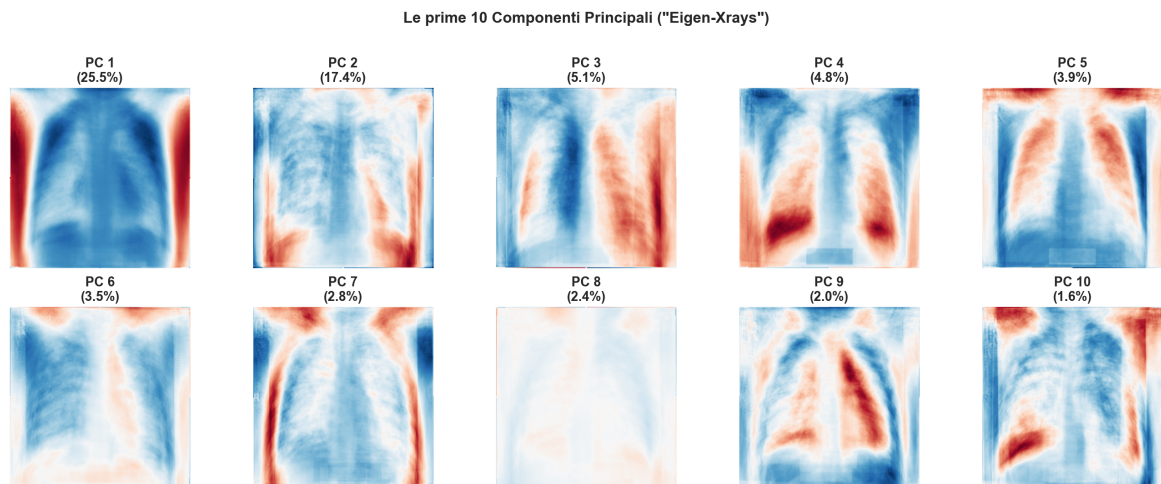


Figure 8: Phase 5 — The first 10 principal components visualised as 256×256 images (RdBu colormap). Red = positive contribution; blue = negative contribution.

Component	Variance	Clinical interpretation
PC1	25.6%	Global brightness contrast — separates bright from dark X-rays
PC2	17.4%	Lateral thoracic symmetry — left vs. right lung profile
PC3	5.1%	Mediastinal pattern — central zone variations
PC4	4.8%	Rib cage structures — costal grid pattern
PC5–10	1.6–3.9%	Fine details: lung fields, diaphragm, apices

Table 5: Clinical interpretation of the first 10 principal components.

5 Comments and Analysis of Results

5.1 SVD for Compression — Key Findings

The Eckart–Young theorem guarantees theoretical optimality; the experiments confirm it empirically. The critical operating point is $k = 20$:

- **Compression ratio:** 15.7% — only $\sim 1/6$ of the original data.
- **PSNR:** 32.4 dB — above the conventional 30 dB diagnostic threshold.
- **Behaviour is class-agnostic:** all four pathologies reach acceptable quality at the same k .

5.2 PCA for Exploratory Analysis — Key Findings

1. **Low-rank structure is universal.** 90% of image variance is encoded by 1–3 singular values across all classes, regardless of pathology. This reflects the high spatial correlation inherent to chest radiographs.
2. **Eigen-X-rays reveal clinically interpretable patterns.** The algorithm “discovers” global contrast (PC1), bilateral symmetry (PC2), and mediastinal structure (PC3) purely from data — without any anatomical prior.
3. **Partial class separability exists.** The COVID-19/Tuberculosis group is separated from Normal/Pneumonia along PC1, but intra-group overlap remains. PCA alone is insufficient for reliable diagnosis.

5.3 Limitations

Imbalanced dataset. Normal dominates the class distribution ($\times 4$ vs. COVID-19), potentially biasing PCA components.

Heterogeneous resolution. Aggressive downscaling (e.g., $1678 \rightarrow 256$ px for Normal) discards fine detail that may be diagnostically relevant.

Pixel-wise metrics. MSE and PSNR do not measure perceptual or clinical quality directly.

Single-image SVD. The SVD analysis characterises one sample per class; variability within a class is not captured.

Unsupervised PCA. PCA maximises global variance, not inter-class discriminability. Supervised alternatives (LDA, kernel PCA) may yield better separation.

5.4 Answers to the Research Questions

1. **Can SVD compress chest X-rays without diagnostic loss?**
Yes. At $k = 20$ (15.7% storage), $\text{PSNR} > 32$ dB. At $k = 50$ (39.1%), $\text{PSNR} > 40$ dB. The Eckart–Young theorem guarantees this is the best possible rank- k approximation.

2. Can PCA distinguish pulmonary pathologies?

Partially. The projection onto PC1–PC2 separates COVID-19/Tuberculosis from Normal/Pneumonia, but with significant overlap. SVD and PCA are powerful *exploratory* tools that support — but do not replace — supervised classifiers for clinical diagnosis.

5.5 Possible Extensions

- **JPEG comparison.** At equal compression ratio, is SVD competitive with the cosine-based JPEG standard?
- **Spatial error map.** Visualise where reconstruction error concentrates anatomically (background vs. lesion regions).
- **t-SNE.** Non-linear dimensionality reduction may reveal sharper inter-class boundaries.
- **Supervised reduction.** Linear Discriminant Analysis (LDA) directly maximises class separability.

Conclusions

Result	Evidence
SVD compresses X-rays effectively	$k = 20$ (15.7% data) $\Rightarrow$ PSNR > 32 dB
Radiographs have strong low-rank structure	90% variance in $k \approx 1$ –3 components
Eckart–Young guarantees optimality	No rank- k method can do better
PCA $\equiv$ SVD on centred data	Same mathematical foundation, complementary goals
Eigen-X-rays reveal clinical patterns	PC1: contrast, PC2: symmetry, PC3: mediastinum
Partial class separability via PCA	COVID-19/TB vs. Normal/Pneumonia separated on PC1
Pre-processing is critical	Tilt correction and image filtering improve PCA quality